



Qualified for the RoboCup World Finals after a successful run at the RCJ Egypt competition at AAST in Alamein, Egypt.

Pharaoh EGYPT - Rescue Line



Team Members during the closing ceremony for RCJ Egypt April 2026

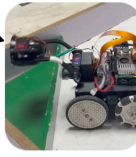
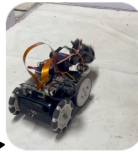
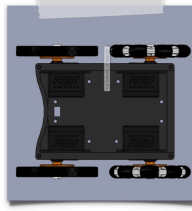
Mechanical Systems



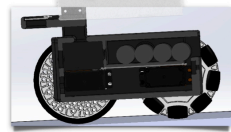
Motor and Wheels configuration

4 REV Smart Servo "Used for motion"
Built with metal gearing delivers rapid, precise movements for nimble control

REV DUO Motion Wheels
Provide strong grip, smooth movement, and precise maneuverability, reducing collisions with the map.



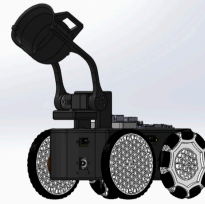
Victim Catching Mechanism use 2 Micro Servo
The mechanism utilizes the high-torque micro servos to ensure swift and secure retrieval of victims within the evacuation zone.



Custom 3D Battery Holder CAD

3D Printed Battery Holder

To ensure electrical stability and reliable power delivery during complex maneuvers, we custom-designed and 3D printed a secure battery holder for the Panasonic NCR18650 cells. This keeps the batteries tightly secured inside the chassis, preventing power disconnections when navigating rough terrain or traversing speed bumps.

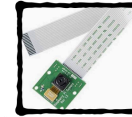


Robot CAD Design using SolidWorks

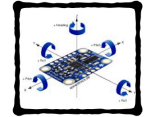
Hardware System & Sensors



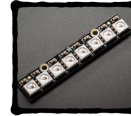
Raspberry Pi 5 Main computer
responsible for robot control, computer vision, AI processing, and high-level decision making.



Dual Raspberry Pi Cameras: One camera tracks the line, while the second camera performs victim detection and evacuation zone analysis.



BNO055 IMU Sensor: Provides orientation and heading data for stable navigation and accurate robot positioning.



NeoPixel LED Strip
Supplies controlled illumination for reliable camera vision under varying lighting conditions.

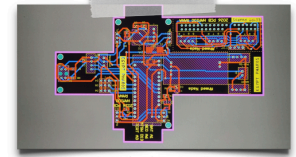


Google Coral USB Accelerator: Accelerates YOLO AI inference, enabling real-time victim detection without overloading the Raspberry Pi.



Yahboom TinyF Laser Ranging Module: Measures precise distances for obstacle detection, wall tracking, and evacuation zone navigation.

Custom Power & PCB Integration
Created in Altium Designer to reduce messy wiring, save space, and improve electrical stability which lead us to reduce wiring by 60%.



Main Control and Power PCB Board

Software

Distributed Computing Architecture

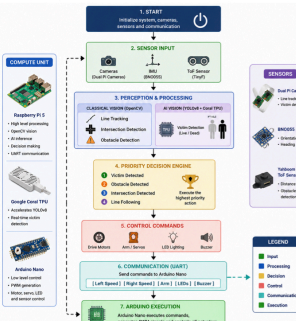
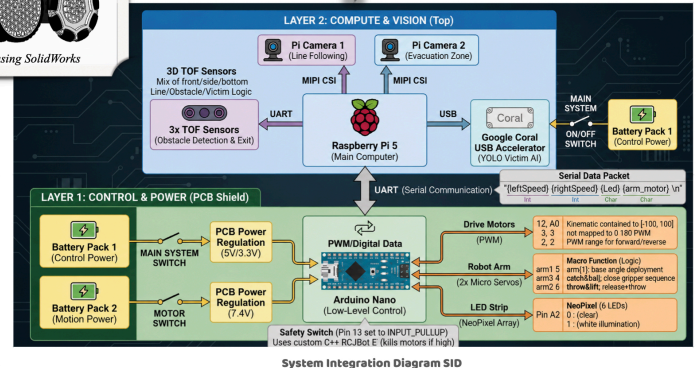
Our robot uses a distributed processing architecture that separates high-level perception from low-level motion control.

Raspberry Pi 5 (Python)

- Captures camera images at 90 FPS
- Processes line, obstacle, and intersection detection using OpenCV
- Executes YOLOv8 victim detection
- Runs the mission state machine and decision logic
- Sends movement commands through UART

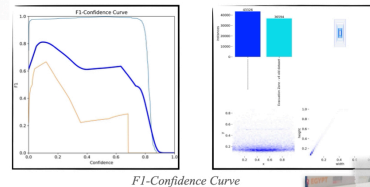
Arduino Nano (C++)

- Receives motor and arm commands
- Generates precise PWM signals
- Controls drive wheels, servos, LEDs, and safety systems
- Executes commands with deterministic timing



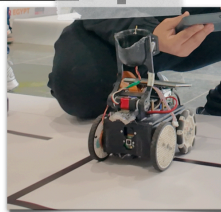
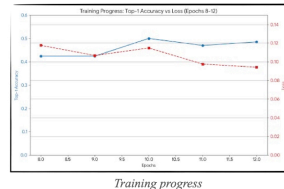
Algorithm, Testing & Validation

Custom YOLOv8 + Coral TPU classifies live (silver) and dead (black) victims. The model was trained on 43,326 images and validated on 36,594 images, achieving a peak F1-score of 0.81. Real-world testing confirmed reliable victim detection and autonomous retrieval under competition conditions.



System Performance

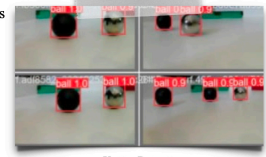
Metric	Value
Camera FPS	90
AI Classes	2
F1 Score	0.81
Training Images	43,326
Validation Images	36,594
Processor	Raspberry Pi 5
AI Accelerator	Coral TPU
Communication	UART



Robot in Action at RCJ Egypt



Pi Screen on the robot

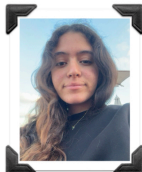


Victim Detection

LCD Touchscreen Interface

Integrated above the victim storage system to provide an intuitive interface for debugging and monitoring sensor data without a keyboard.

Team Members



Maya Ragab
(Main Programmer)

Developed the robot software architecture, AI vision system, and hardware integration.



Diala Ragab
(Second Lead Programmer)

Contributed to software development, system integration, debugging, and performance optimization.



Ruba Ahmed
(Electrical & Custom PCB Designer)

Designed the custom PCB and power distribution system supporting reliable robot operation.



Youssef Mohamed
Main Mechanical Designer

Designed the robot chassis, victim retrieval mechanism, and custom 3D-printed components.